

RECAP: Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

Consider a Time Homogeneous DTMC $\{X_n : n \geq 0\}$ on the state space S with transition probability matrix P . $P_{ij} = \mathbb{P}\{X_{n+1} = j | X_n = i\}$.

1) Stopping time - definition

2) Strong Markov property of a TH DTMC.

3) Hitting Times and Recurrence k^{th} hitting time of state $j = T_j^k$
 - First passage time distribution: $f_{ij}^{(n)} = \mathbb{P}(T_j^{(1)} = n | X_0 = i)$
 - Mean recurrence time: $v_{jj} = \sum_{n=1}^{\infty} n \cdot f_{jj}^{(n)}$

Consider a TH DTMC $\{X_n, n \geq 0\}$ on a state space S , defined w.r.t $(\Omega, \mathcal{F}, \mathbb{P})$.

For $i, j \in S$, $f_{ij}^{(n)} = \mathbb{P}\{X_1 \neq j, X_2 \neq j, \dots, X_{n-1} \neq j, X_n = j | X_0 = i\}$ prob. that first visit to state j is at time n when process starts in i
 $f_{ij} = \sum_{n=1}^{\infty} f_{ij}^{(n)} \rightarrow$ prob. of ever hitting j when process starts off in i .

$f_{ij}^{(n)}$ is a pmf on $\mathbb{N}$, can be defective, in general.

First passage time distribution for hitting j starting from i .

$$f_{ij} < 1$$

$$\Rightarrow (1 - f_{ij}) > 0.$$

$f_{jj}^{(n)}$ - prob. of first return to state j at time n

If $f_{jj} = 1$, Mean recurrence time $v_{jj} = \sum_{n=1}^{\infty} n f_{jj}^{(n)}$

PLAN: 1) Convergence of Expectation & Monotone Convergence Theorem

2) Transient and Recurrent States

3) Communicating Classes and Class Properties

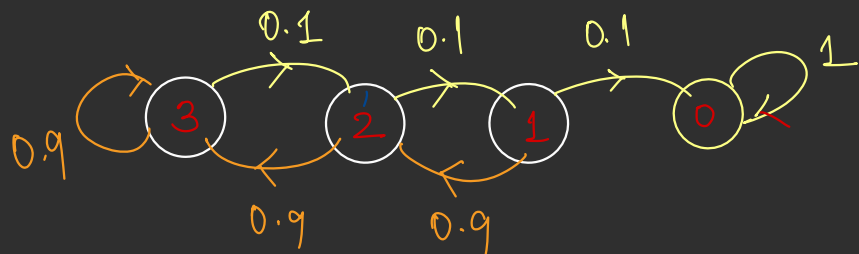
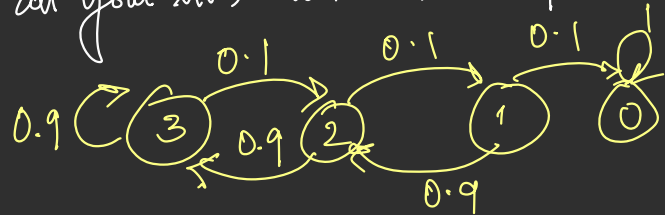
4) Positive Recurrence & Stationary Distribution

REFERENCE: 1) A. Kumar, "Discrete Event Stochastic Processes"

2) Lecture Notes of the course "Stochastic Processes", by Dr. P.N. Karthik, IIT.

Goldmann Sachs Interview Question @ IISc (2017-18 Placement Season)

Imagine that you are playing a game that involves tossing a coin of bias 0.9 repeatedly and independently, with 3 lives at the start of the game. At each round, if the coin toss results in a tail, you lose one life. However, if the coin toss results in a head, you gain back your lost lives one at a time. If you have all 3 lives and the coin toss results in a head, nothing changes and you continue tossing. You are allowed to play until you lose all your lives. What is the expected number of times you will toss the coin?



$$\mathbb{P}\{X_n = 0 \mid X_0 = 3\}$$
$$f_{30}^{(n)} \quad \mathbb{E}[T_0^1 \mid X_0 = 3]$$

n_i - the average number of tosses to go from i to 0 $i \in \{1, 2, 3\}$.

$$n_0 = 0$$

$$n_1 = 0.1(1 + n_0) + 0.9(1 + n_2)$$

$$n_2 = 0.1(1 + n_1) + 0.9(1 + n_3)$$

$$n_3 = 0.1(1 + n_2) + 0.9(1 + n_3)$$

Required to find n_3 .

You find $f_{30}^{(n)}$.

$$n_3 = \mathbb{E}[T_0^1 \mid X_0 = 3]$$

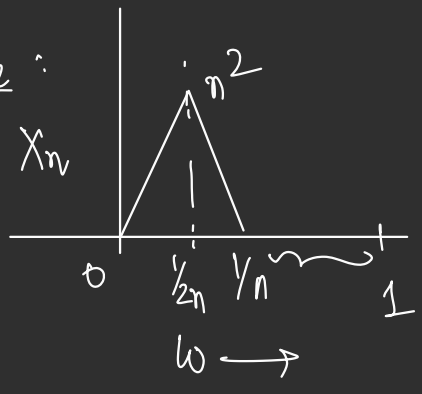
Convergence of Expectation - $(\Omega, \mathcal{F}, \mathbb{P})$

If $X_1, X_2, \dots$ is a sequence of RVs converging almost surely to the RV X

($X_n \xrightarrow{a.s.} X$). Is it true that $\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[\lim_{n \rightarrow \infty} X_n]$?

i.e. if $X_n \xrightarrow{a.s.} X$, is $\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X]$?

Example:



$$\Omega = [0,1], \mathcal{F} = \mathcal{B}([0,1]), P = \text{unif}[0,1]$$

for each $\omega \in \Omega \setminus \{0\}$, $\lim_{n \rightarrow \infty} X_n(\omega) = 0$

$$X_n \xrightarrow{\text{a.s.}} X$$

$$X(\omega) = 0 \quad \forall \omega \in \Omega$$

$$E[X_n] = n/2$$

$$E[X] = 0$$

$$\lim_{n \rightarrow \infty} E[X_n] \neq E\left[\lim_{n \rightarrow \infty} X_n\right]$$

The following result provides conditions that ensure the validity of the exchange of expectation & almost-sure limit of random variables.

Monotone-Convergence Theorem: If $X_n, n \geq 1$ is a seq. of random variables

that converges almost surely to the RV X , i.e., $X_n \xrightarrow{\text{a.s.}} X$, and the following two conditions hold:

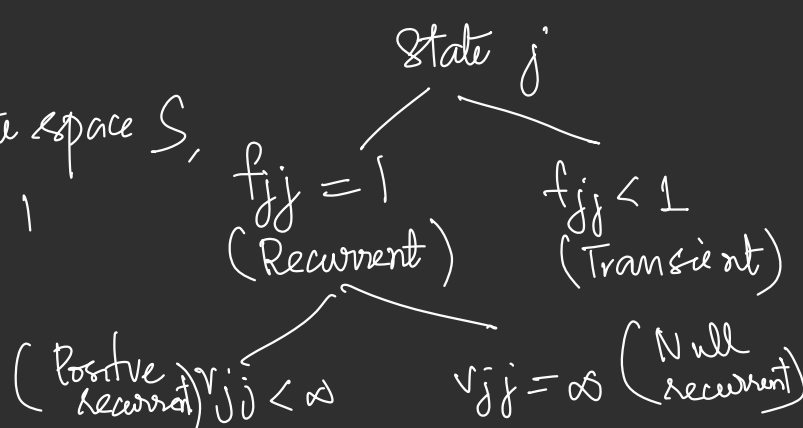
- i) X_n is non-negative for every $n \geq 1$ and
- ii) $X_1 \leq X_2 \leq \dots \leq X_n \leq \dots$ (seq. of monotonically non-decreasing RVs).

$$\forall \omega \in \Omega \quad X_1(\omega) \leq X_2(\omega) \leq \dots \leq X_n(\omega) \leq \dots$$

then $\lim_{n \rightarrow \infty} E[X_n] = E[X]$ { $E[X] = \infty$ is a possibility here! }

Transient and Recurrent States

For a TH-DTMC $\{X_n\}_{n \geq 0}$ on a state space S ,
 a state $j \in S$ is called transient if $f_{jj} < 1$
 recurrent if $f_{jj} = 1$



Positive recurrent if $f_{jj} = 1$ & $\nu_{jj} < \infty$

Null recurrent if $f_{jj} = 1$ & $\nu_{jj} = \infty$.

Number of Returns to a State

Let $M_j = \sum_{n=1}^{\infty} \mathbb{I}_{\{X_n=j\}}$ - Number of visits to state j for $n \geq 1$.

$$\mathbb{E}[M_j | X_0=j] = \mathbb{E}\left(\sum_{n=1}^{\infty} \mathbb{I}_{\{X_n=j\}} | X_0=j\right). \quad \mathbb{P}\{X_n=j | X_0=j\}$$

$$= \sum_{n=1}^{\infty} (P^n)_{jj} \quad \left(\text{Uses Monotone Convergence Theorem} \right)$$

Check how!

Theorem

For a TH-DTMC $\{X_n\}_{n \geq 0}$ on the state space S if a state j is **transient** then for $m = \{0, 1, 2, \dots\}$

$$\mathbb{P}(M_j = m | X_0=j) = f_{jj}^m (1 - f_{jj})$$

Proof: Homework!

If j is recurrent, $\mathbb{P}(M_j = m | X_0=j) = 0$ for $m \in \mathbb{N}$

Using this result, if j is transient,

$$\mathbb{E}[M_j | X_0=j] = \sum_{m=1}^{\infty} m f_{jj}^m (1 - f_{jj}) = \frac{f_{jj}}{1 - f_{jj}} < \infty$$

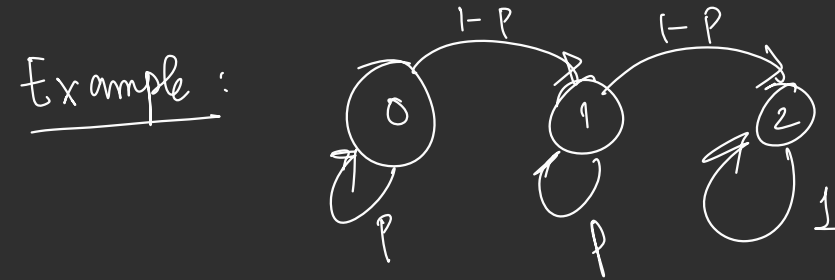
And if it is recurrent $\mathbb{E}[M_j | X_0=j] = \infty$.

Thm: A state j is transient if the expected number of returns is finite
Thus, for a recurrent state j , the expected number of returns is infinite

$$\sum_{n=1}^{\infty} (P^n)_{jj} = \infty$$

$$\sum_{n=1}^{\infty} (P^n)_{jj} < \infty$$

For a transient state j , $\lim_{n \rightarrow \infty} (P^n)_{jj} = 0$



$$V_{22} = \sum_{n=1}^{\infty} n \cdot f_{22}^{(n)} = 1 < \infty$$

Positive Recurrent

$$\begin{aligned} f_{00} &= p \\ f_{11} &= p \\ f_{22} &= 1 \end{aligned} \left. \vphantom{\begin{aligned} f_{00} &= p \\ f_{11} &= p \\ f_{22} &= 1 \end{aligned}} \right\} \begin{array}{l} \text{Transient} \\ \text{Transient} \\ \text{Recurrent} \end{array}$$

$$\begin{cases} f_{22}^{(1)} = 1 \\ f_{22}^{(n)} = 0 \quad \forall n > 1 \end{cases}$$

$$f_{jj}^{(n)} = P(X_1 \neq j, \dots, X_{n-1} \neq j, X_n = j | X_0 = j)$$

Communicating Classes & Irreducibility of a DTMC.

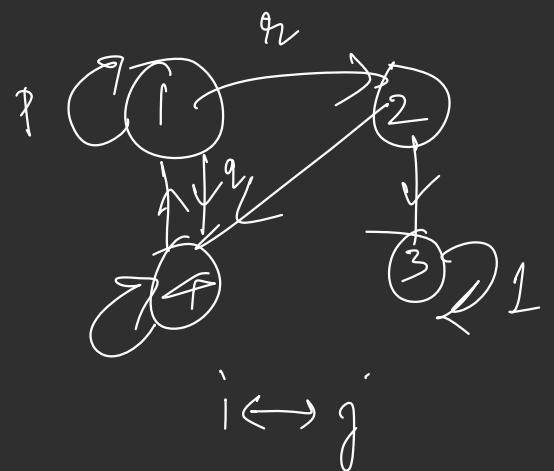
Defn: Given a TH-DTMC $\{X_n, n \geq 0\}$ on the state space S , for $i, j \in S$, we say that

i) j is reachable from i if $\exists n \geq 0$ s.t.

$$P_{ij}^{(n)} > 0 \quad (\text{Notation: } i \rightarrow j)$$

($\exists n \geq 0$ s.t. j can be reached from i in n steps with positive prob.)

ii) i & j communicate if $i \rightarrow j$ & $j \rightarrow i$

$$(i \leftrightarrow j)$$


8 am to 9 am
9 am to 10 pm
1 pm
2 to 5

Prove that $\longleftrightarrow$ is an equivalence relation on S .

i) Reflexive: By definition $P_{jj}^0 = 1$

ii) Symmetry

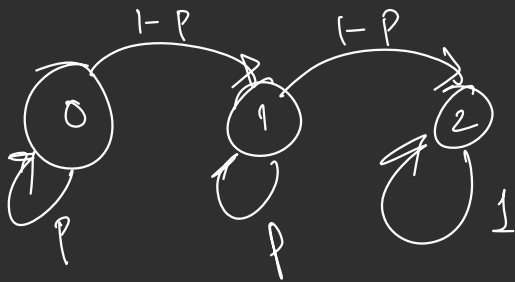
iii) Transitive

$$i \longleftrightarrow j, j \longleftrightarrow k \Rightarrow i \longleftrightarrow k$$

$$\begin{array}{ll} i \longleftrightarrow j & j \longleftrightarrow k \\ i \xrightarrow{n_1} j & j \xrightarrow{m_1} k \\ j \xrightarrow{n_2} i & k \xrightarrow{m_2} j \end{array}$$

$\longleftrightarrow$ partitions S into equivalence classes.

Example:



C_1, C_2, C_3
 $\{0\}, \{1\}, \{2\}$ — closed
 OPEN.

Defn: Given a TH-DTMC $\{X_n, n \geq 0\}$ on the state space S with TPM P

i) The classes into which $\longleftrightarrow$ partitions S are called communicating classes.

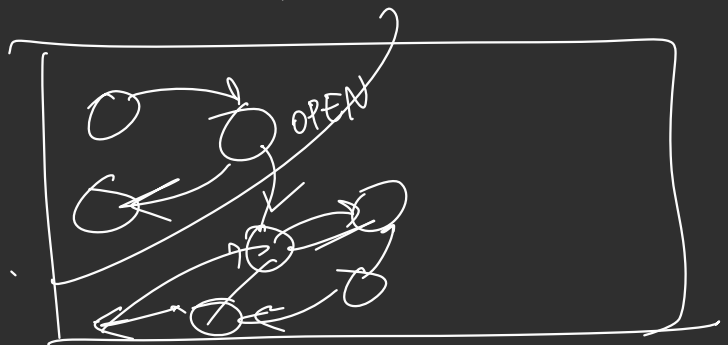
ii) A communicating class C is said to be closed if $\forall i \in C \ \& \ j \notin C, P_{ij} = 0$

Otherwise called OPEN

iii) The Markov Chain is said to be **IRREDUCIBLE** if the entire S is one communicating class.

An irreducible DTMC is closed.

(Irreducibility / Unichain property).



Class Property Theorem: The states in a communicating class are either

- i) all transient
- ii) all null-recurrent (OR)
- iii) all positive-recurrent

} Transience and Recurrence are class properties

Invariant & Stationary Distribution (steady state or Equilibrium distribution)

Consider a TH DTMC on state space S with TPM P .

A PMF π on S is called an invariant distribution for P if $\pi = \pi P$

$$\text{i.e. } \forall j \in S \quad \pi_j = \sum_{i \in S} \pi_i P_{ij}$$

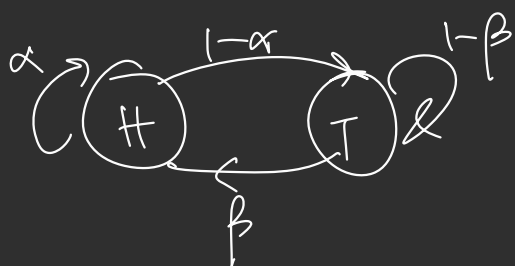
The equation $\pi = \pi P$ is called the global balance equation is a system of $|S|$ linear equations

π is a left-eigen vector of P with eigen value 1.

From Chapman-Kolmogorov Equation $\pi(n+1) = \pi(1) \cdot P^n$ for any $n \in \mathbb{N}$

If $\pi(1) = \pi$ (an invariant distribution) $\pi(n) = \pi$ for any $n \in \mathbb{N}$

This together with time-homogeneity implies that the Markov Chain is stationary



Find an invariant distribution of the two-state DTMC

The invariant distribution may not be unique

$$\pi_1 = \pi_1 P \quad \pi_2 = \pi_2 P.$$

$$\pi_\alpha = \alpha \pi_1 + (1-\alpha) \pi_2 \quad \text{for } \alpha \in [0, 1].$$

$$\pi_\alpha = \pi_\alpha P.$$

Consider $P = I$

then $\pi P = \pi \neq \pi$.

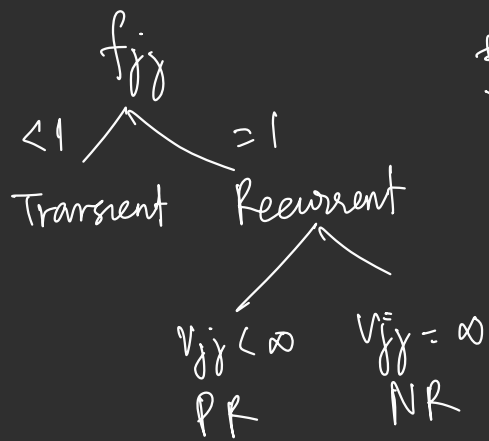
$\pi(j)$ for j transient?
 $= 0$

On existence & uniqueness of invariant distribution.

Let $\{X_n\}_{n \geq 0}$ be an irreducible TH-DTMC on S with TPM P . Then

a unique stationary distribution π exists if and only if $\{X_n\}_{n \geq 0}$ is positive-recurrent

In this case $\pi_j^- = \frac{1}{v_{jj}} > 0 \quad \forall j \in S$. v_{jj} - mean recurrence time
 For a PR state j , $v_{jj} < \infty$.

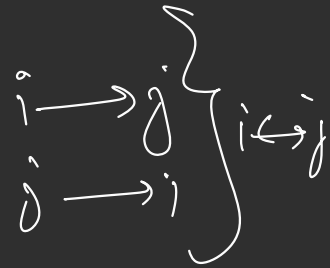


returns to a state
Expected

i communicates with j

Communicating classes

open closed



Class properties: Transient
 (OR) PR
 (OR) NR

Irreducible DTMC: S is one comm. class.

Invariant distribution $\pi = \pi P$.

For an irreducible PR DTMC $\pi_j^- = \frac{1}{v_{jj}}$
 $\forall j \in S$

$\pi_j^- = 0$ for a transient state.

$$\pi(n+1) = \pi(n) P$$

$$\lim_{n \rightarrow \infty} \pi(n+1) = \underline{\underline{\pi}}$$

$$\lim_{n \rightarrow \infty} (P^n)_{jj} = 0$$